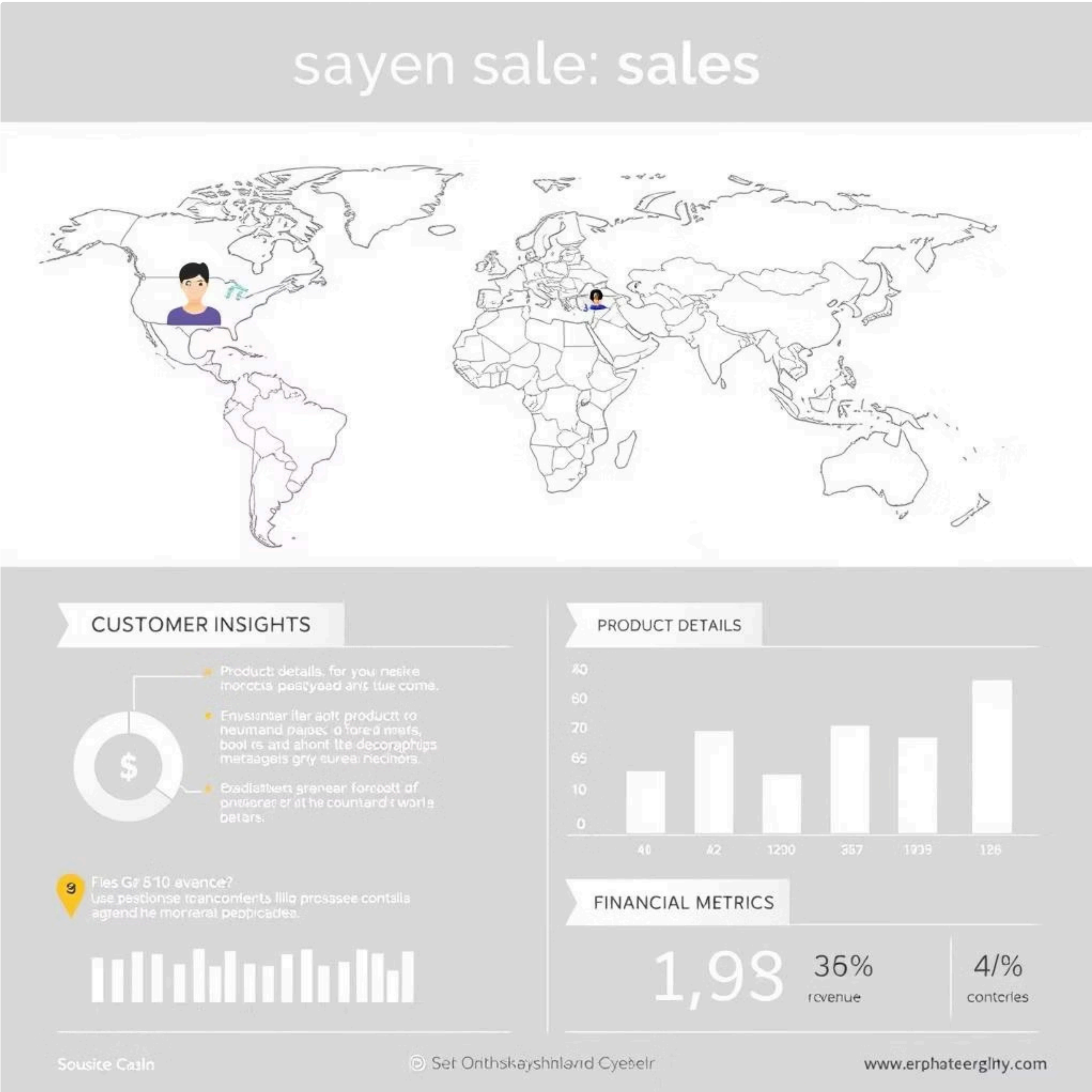
International Sales Report Analysis

This document outlines a Python-based data analytics project focused on deriving actionable business intelligence from raw international sales data. Utilising libraries like Pandas, Matplotlib, and Seaborn, the project performs data cleaning, processing, visualisation, and interpretation to provide dynamic, data-backed insights for optimising sales strategies.into the business context, the challenges, and the project's strategic goals.

In the current digital era, the global marketplace is more dynamic and competitive than ever. For any business operating internationally, daily transactions generate a deluge of data—a valuable but often overwhelming asset. The challenge lies not in collecting this data, but in effectively interpreting it to make swift, intelligent decisions. Manually analyzing thousands of sales records is not only time-consuming and prone to errors but also causes significant delays in responding to market changes. This reactive approach can lead to missed sales opportunities, inefficient inventory management, and poorly targeted marketing efforts, ultimately eroding profitability and competitive standing.

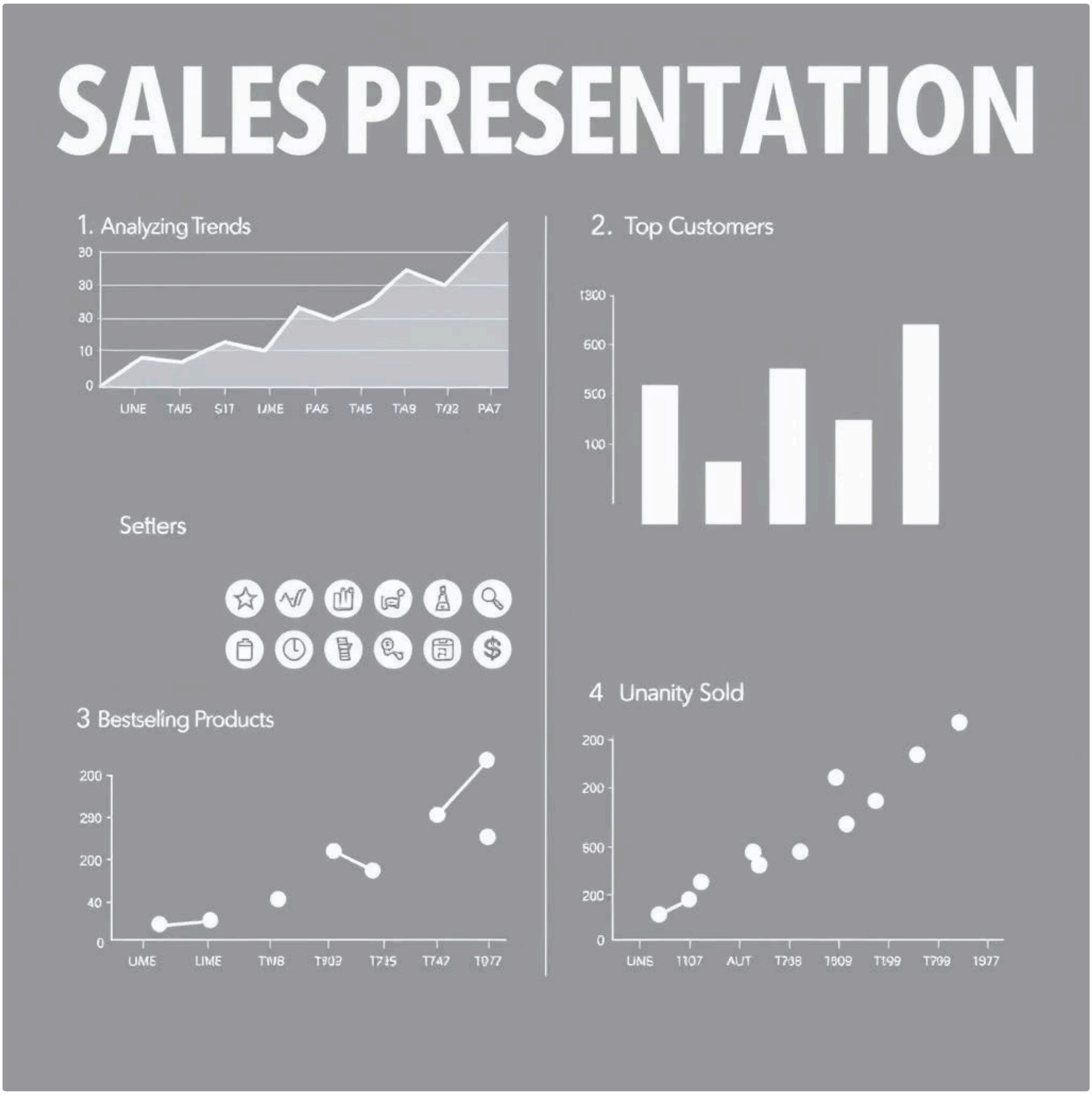
1. Abstract

The International Sales Report Analysis project leverages Python and robust data analytics libraries to extract actionable business intelligence from raw sales data. It processes over 17,000 transactional records, including customer name, style code, SKU, date, size, rate, and gross amount. Through Pandas, Matplotlib, and Seaborn, the project conducts data cleaning, processing, visualisation, and interpretation, offering a dynamic, data-backed perspective to understand revenue drivers and optimise sales strategies.



2. Objectives

- Clean and preprocess raw international sales data.
- Analyse sales trends over time.
- Identify top customers by revenue.
- Highlight bestselling product styles and sizes.
- Explore the relationship between quantity sold and revenue.
- Visualise and summarise findings in an actionable format.



3. Dataset Description

Column	Description
index	Row index
DATE	Date of transaction
Months	Sales month
CUSTOMER	Customer name
Style	Product style code
SKU	Stock Keeping Unit
Size	Size (e.g., L, XL, XXL)
PCS	Quantity sold
RATE	Unit price
GROSS AMT	Total sale amount

Column Name	Description
`index`	A unique identifier for each transaction row (auto-incremented).
`DATE`	The exact date when the sale occurred, in MM-DD-YY format.
`Months`	A shorthand label for the sales month (e.g., "Jun-21"). Useful for grouping and monthly analysis.
`CUSTOMER`	The name of the customer who made the purchase. This can be used to analyze customer-wise sales trends.
`Style`	A code representing the design or product type sold. Helps identify top-perfor[illegible] styles.

4. Tools & Technologies Used

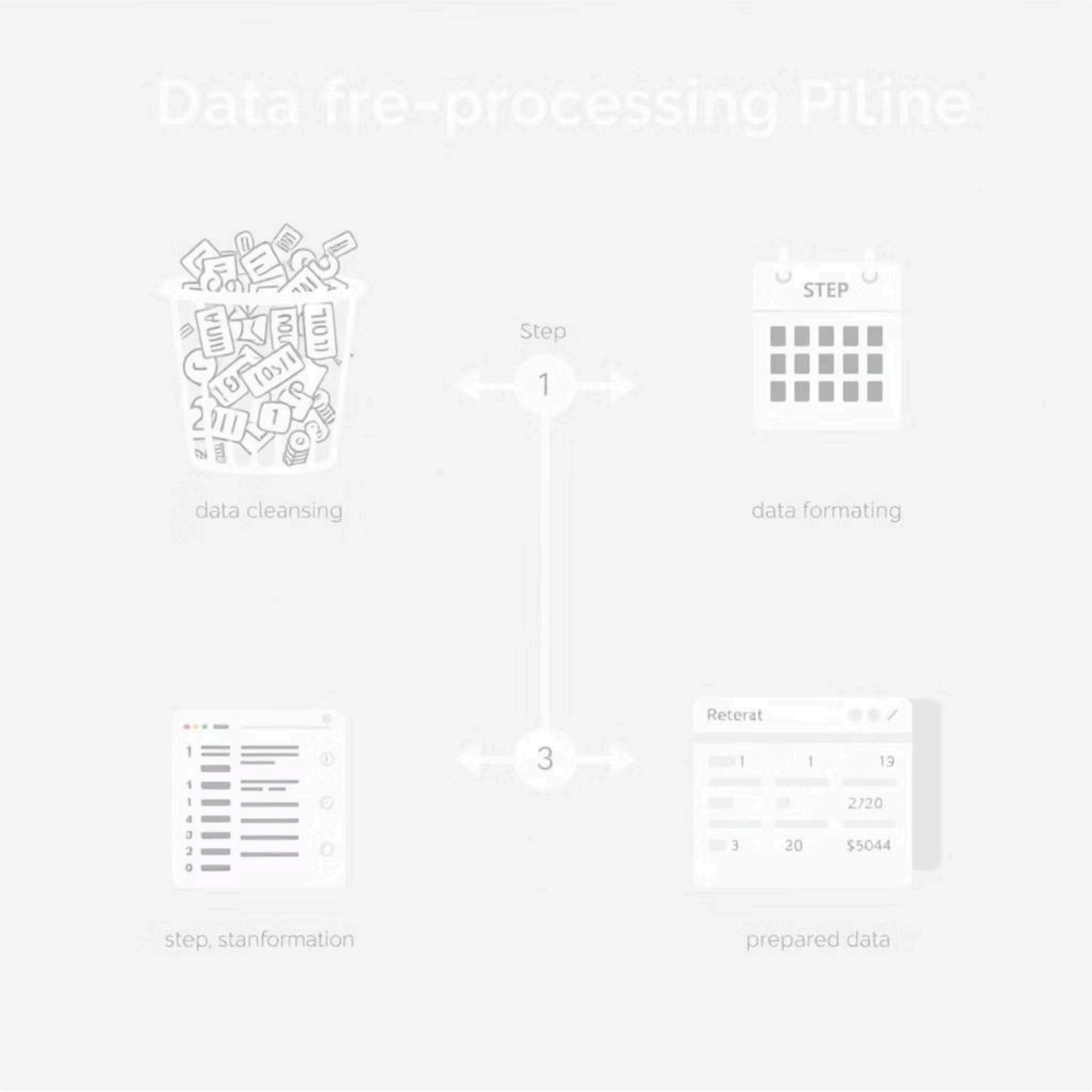
- **OS:** Windows/macOS/Linux
- **Language:** Python 3.8+
- **IDE:** Jupyter, VS Code, or Anaconda
- **Libraries:** pandas, matplotlib, seaborn
- **Data Format:** .csv (Excel-compatiable)

5. Data Preprocessing

Data preprocessing involved several crucial steps to ensure data quality and readiness for analysis:

- Removed duplicate and null values to maintain data integrity.
- Converted the 'DATE' column to datetime format for time-series analysis.
- Transformed 'PCS', 'RATE', and 'GROSS AMT' columns to float data types for numerical operations.

The final cleaned dataset comprises **17,218 rows × 10 columns**, providing a robust foundation for subsequent analysis.



Explanation of Python Libraries Used

Pandas: Data Manipulation

Crucial for handling sales data, pandas enabled efficient loading, cleaning, and transformation of the raw dataset into a structured format for analysis.

Matplotlib: Core Visualisation

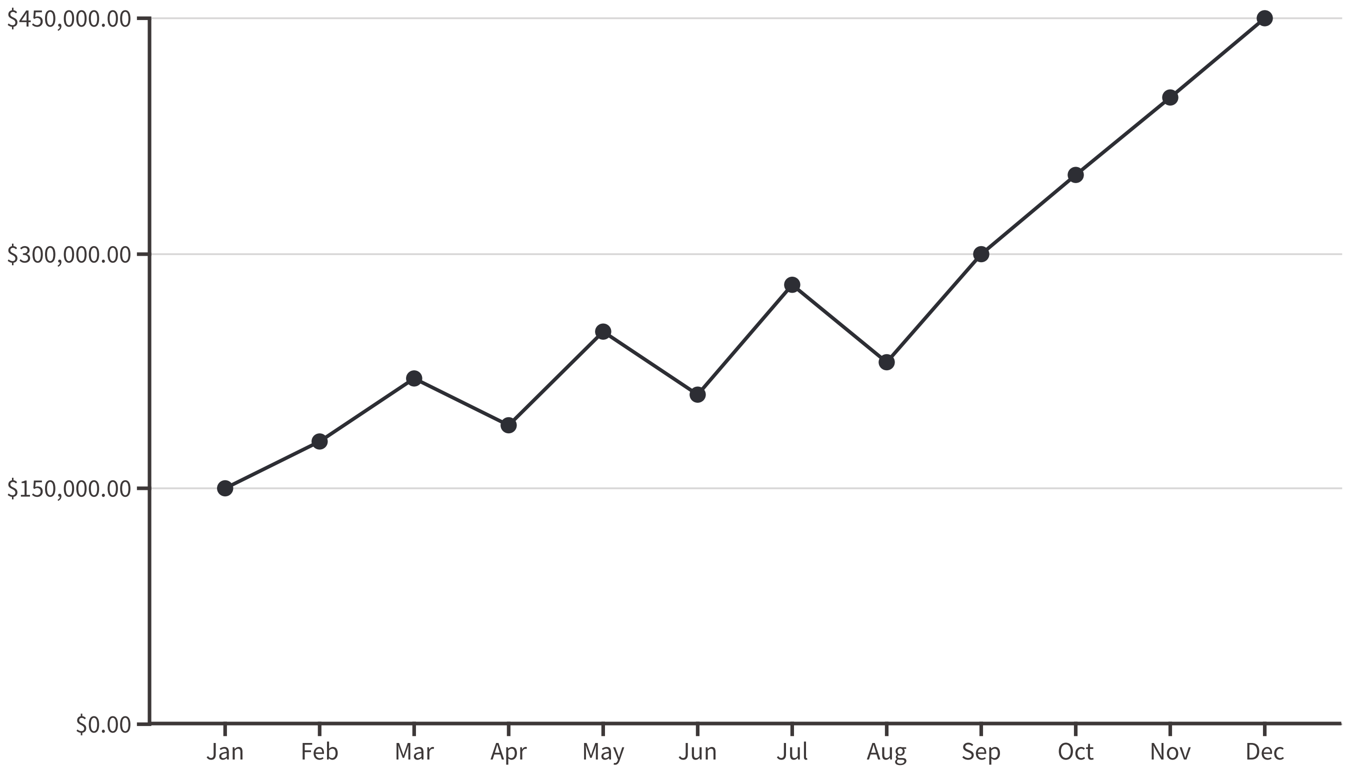
This library provided the foundational tools for creating diverse static charts, enabling clear visual representation of sales trends and patterns over time.

Seaborn: Enhanced Statistical Plots

Built on matplotlib, seaborn was instrumental for generating more sophisticated and aesthetically rich statistical graphics, aiding deeper insights into sales distributions.

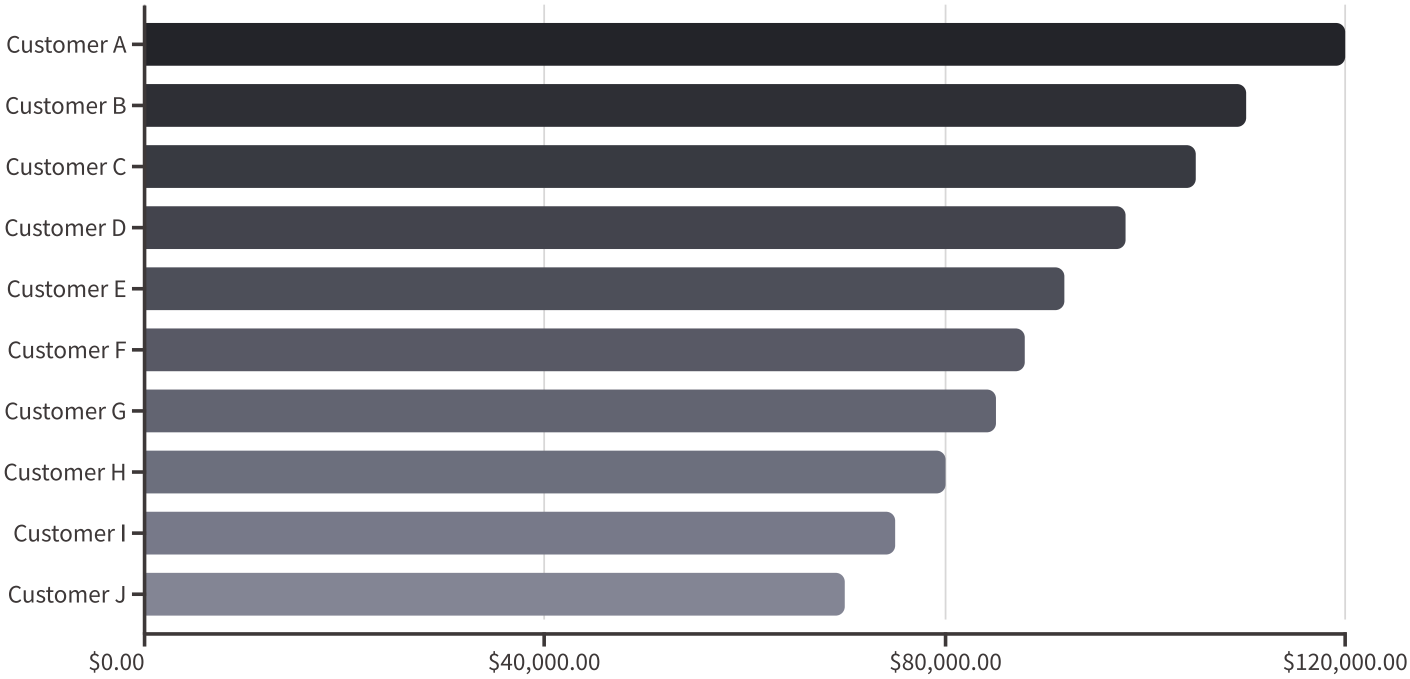
6. Gross Sales Over Time

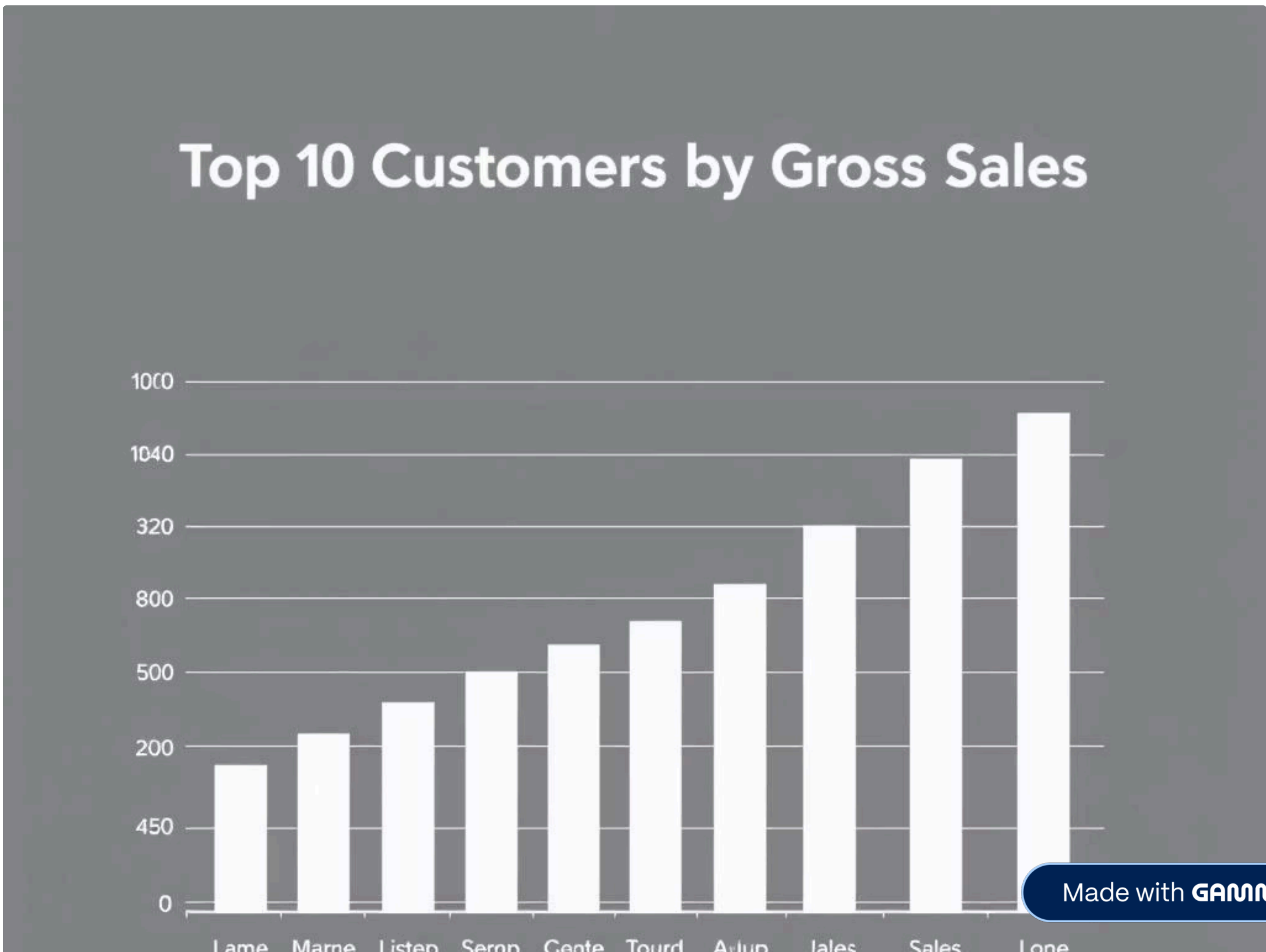
This line chart illustrates the fluctuation of revenue over time. Notable spikes in certain months suggest the impact of marketing campaigns or festive seasons, indicating periods of increased sales activity.



7. Top 10 Customers by Gross Sales

This chart identifies the top 10 customers who contribute the highest revenue. These high-value customers are ideal candidates for targeted loyalty programmes and exclusive offers, helping to foster stronger relationships and ensure continued business growth.



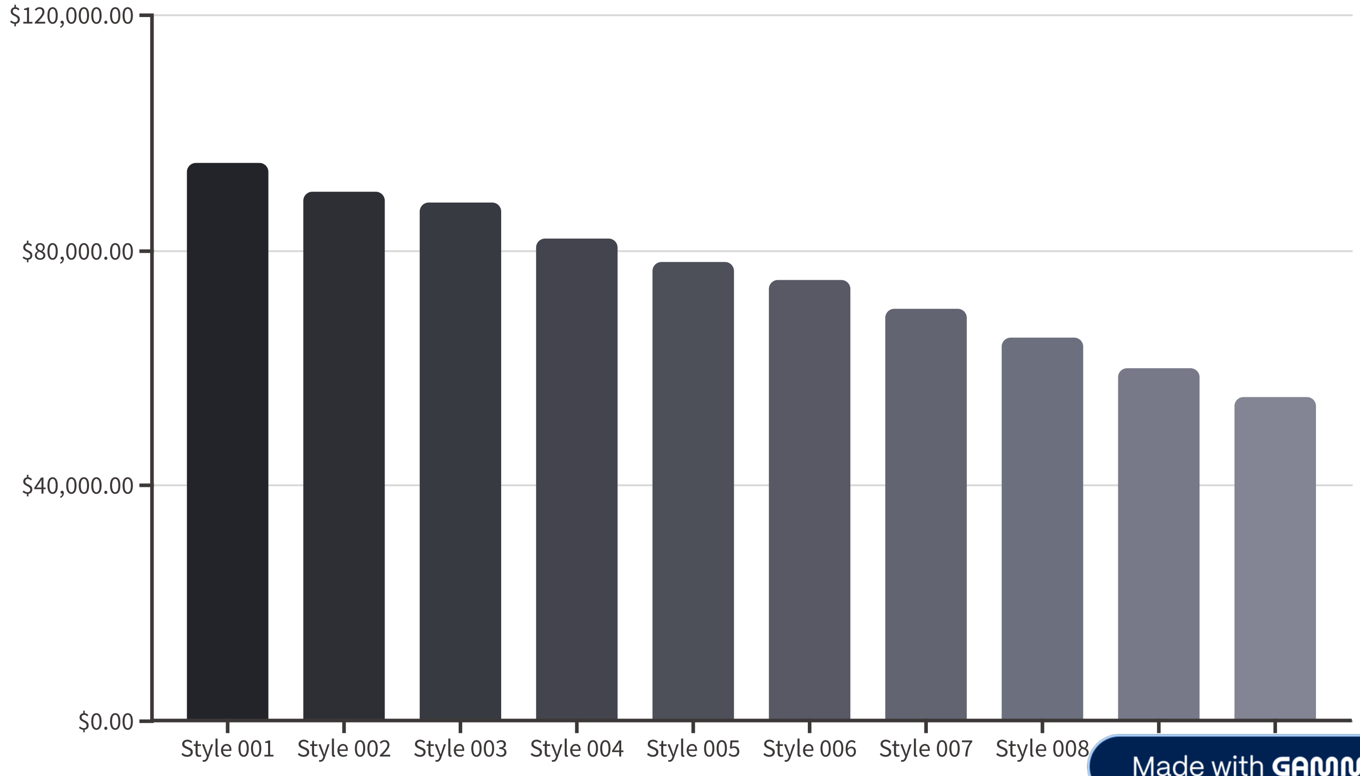


8. Top 10 Styles by Gross Sales

This chart highlights the best-performing products based on their style codes. Identifying these top styles is crucial for inventory management, marketing focus, and future product development, ensuring resources are allocated effectively to maximise sales and profitability.

🔍 Key Insights:

- **Revenue Concentration**:
  A small number of styles contribute a disproportionately large share of total sales revenue. This is a classic example of the **Pareto Principle (80/20 Rule)** — where roughly 20% of products (styles) generate 80% of the income.
- **Customer Preference**:
  The styles at the top reflect current market demand and customer taste. These could be linked to specific fabric types, fits, patterns, or seasonal trends.
- **Production Planning**:
  Understanding which styles consistently perform well enables better forecasting and planning. Manufacturers can allocate more production resources to high-demand styles while phasing out underperformers.
- **Marketing Focus**:
  Popular styles should be highlighted in promotional campaigns, featured prominently on websites/catalogs, and possibly expanded with new color or size variants.
- **Product Lifecycle Management**:
  Styles that appear regularly across months may indicate **core items** with long shelf-life, while others may be **trending or seasonal**.

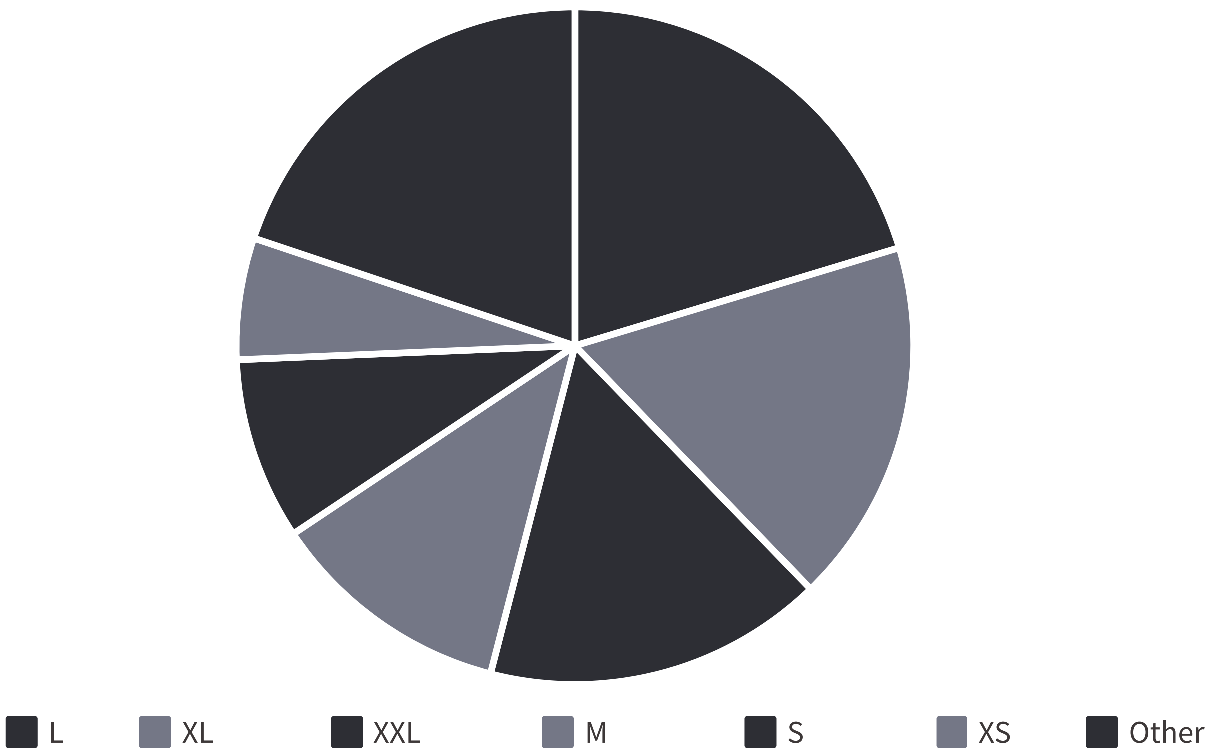


9. Size Distribution

This chart illustrates the popularity of different product sizes, with **L**, **XL**, and **XXL** being the most frequent. This insight is invaluable for optimising inventory planning, ensuring adequate stock of popular sizes, and reducing potential overstocking of less popular ones.

🔍 Key Insights:

- **High Demand Sizes**:
  Sizes like **L, XL, and XXL** dominate the sales, indicating they are the preferred fit for a majority of the customer base. This trend may reflect the demographic or target market being served (e.g., adult apparel).
- **Low Demand Sizes**:
  Sizes such as **S, XS, or unique extended sizes (e.g., 3XL, 4XL)** appear less frequently. These may cater to a niche market and should be produced or stocked in smaller quantities to avoid overstock.




Thank You